

B. Sc. (Hons) Semester V Open Book Examination 2021 – 22**PHYSICS****Paper No. BPT-501 : Mathematical Physics****Time : 4 hours 30 minutes****Full Marks : 70****Instructions :**

- (i) The Question Paper contains 08 questions out of which you are required to answer any 04 questions. The question paper is of 70 Marks with each question carrying 17 · 5 Marks.
- (ii) The total duration of the examination will be 4:30 Hours (Four Hours and Thirty Minutes), which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
- (iii) For the students with benchmark disability as per Persons with Disability Act, the total duration of examination shall be 6 Hours (Six Hours) to complete the examination process, which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
- (iv) Answers should be hand-written on plain white A4 size paper using black or blue pen. Each question can be answered in up to 350 words on 3 (Three) plain A4 size paper (only one side to be used).
- (v) Answers to each question should start from a fresh page. All pages are required to be numbered. You should write your Course Name, Semester, Examination Roll Number, Paper Code, Paper Title, Date and Time of Examination on the first sheet used for answers.
- (vi) Unless otherwise specified, the mathematical symbols have their usual meanings.

Q.1. 1. Find the expressions for the quantum mechanical angular momentum operators, L_x , L_y , L_z and L^2 in spherical polar coordinates, starting from $\vec{L} = -i(\vec{r} \times \vec{\nabla})$. **10**

2. Using the expression $\vec{L} = -i(\vec{r} \times \vec{\nabla})$, verify that $\vec{L} \times \vec{L} = i\vec{L}$ in spherical polar coordinates. **$7\frac{1}{2}$**

Q.2. 1. Using index notation and ε_{ijk} , show that for any three vectors, $\vec{u}$, $\vec{v}$ and $\vec{w}$

$$[\vec{u} \times (\vec{v} \times \vec{w})] + [\vec{v} \times (\vec{w} \times \vec{u})] + [\vec{w} \times (\vec{u} \times \vec{v})] = 0$$

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2. Consider a real symmetric matrix with two distinct eigenvalues α_1 and α_2 . Prove that the corresponding eigenvectors are orthogonal to each other. **$5\frac{1}{2}$**

Q.3. 1. Express the vector

$$\vec{P} = \frac{2p \cos \theta}{r^3} \hat{r} + \frac{p \sin \theta}{r^3} \hat{\theta}$$

in Cartesian coordinates (p is a constant). Can this vector be expressed as the gradient of a scalar function? If yes, find the function. 10

2. Give, with brief explanation, an example from Physics, in which one encounters the expression(s) in part (a) of this question. 4

3. Give two examples of physical quantities that are axial vectors. Provide a brief explanation in each case. $3\frac{1}{2}$

Q.4. 1. Find the eigenvalues and (normalized) eigenvectors of the matrix

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

and diagonalize the matrix A . $12\frac{1}{2}$

2. Consider a similarity transformation, $A' = BAB^{-1}$. Show that the eigenvalues of A and A' are the same. 5

Q.5. 1. Solve $xy'' - xy' + y + py = 0$, with p an integer, about the point $x = 1$, using the Frobenius method. Write down the solutions of the above differential equation for $p = 1, 2, 3$. $12\frac{1}{2}$

2. Evaluate :

$$\int_{-\infty}^{\infty} \sin\left(\frac{\pi x}{2}\right) \delta(x^2 - 3x + 2) dx$$

5

Q.6. 1. Using the Rodrigues' formula for Legendre polynomials, show that—

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$$

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2. Make a neat sketch of the periodic function (including at least two periods on each side)

$$f(x) = \begin{cases} \frac{1}{2}(\pi - x), & 0 < x \leq \pi \\ -\frac{1}{2}(\pi + x), & -\pi \leq x < 0 \end{cases}$$

and find its Fourier series. What is the value of $f(x)$ given by the Fourier series? Explain. $9\frac{1}{2}$

Q.7. 1. Using recurrence relations, show that—

$$\int_{-1}^1 x P_n(x) P_{n-1}(x) dx = \begin{cases} \frac{2l}{(2l-1)(2l+1)}, & m = l-1 \\ \frac{2(l+1)}{(2l+1)(2l+3)}, & m = l+1 \end{cases}$$

$9\frac{1}{2}$

2. The Maxwell's speed distribution function is given by

$$\frac{dN}{N} = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left(-\frac{mv^2}{2kT} \right) v^2 dv$$

where the terms have their usual meanings. The average value of v^n is defined as $\langle v^n \rangle = \frac{1}{N} \int v^n dN$. Show that—

$$\langle v^n \rangle = \left(\frac{2kT}{m} \right)^{n/2} \frac{\Gamma((n+3)/2)}{\Gamma(3/2)}$$

(Here $\Gamma(\dots)$ stands for the Gamma function).

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- Q.8.** 1. Make a neat sketch of the function

$$f(x) = \begin{cases} x(\pi - x), & 0 < x < \pi \\ x(\pi + x), & -\pi < x < 0 \end{cases}$$

and find its Fourier series.

$9\frac{1}{2}$

2. How many terms have to be included in this Fourier series to match the maximum value of the function to three significant figures? Show by calculation. Comment on the convergence of the Fourier series.

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External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>